Primary Prevention in British Columbia:
Preliminary Findings on Quantitative Cancer Risk Indicators

Summary Report

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Background

Modifiable lifestyle factors, environmental exposures and infections contribute to the development of many cancer types[1]. Approximately four in ten cancer cases can be prevented through healthy living behaviours and supportive policies [2]. According to the Canadian Population Attributable Risk of Cancer (ComPARe) study conducted in 2015, it is estimated that between 33-37% of incident cancer cases among adults aged 30 years and over were attributable to preventable risk factors (see Appendix A). 8,500 cancer cases were preventable in British Columbia alone[2]. Organizations such as the Canadian Cancer Society (CCS) and the World Cancer Research Fund International (WCRF) have recognized the most common risk factors to be: smoking, alcohol, physical activity, diet, ultraviolet (UV) radiation, environmental exposures, occupational exposures and infections (particularly HPV and hepatitis B)[3,4]. Opportunities to reduce the impact of cancer risk factors exist through implementation of evidence-based policies and programs at the individual, community and societal levels[5]. For example, The Prevention System Quality Index (PSQI) 2020 by Cancer Care Ontario is the 4th report in a series focused on policy indicators addressing cancer risk factors in Ontario[6]. It contains evidence and data intended for usage by policy makers and program planners in cancer prevention activities[6]. It also identifies current activities that have been successful in cancer prevention and possible gaps[6]. Given that the provision of health care is decentralized in Canada, similar resources focused on indicators specific to other provinces and territories may be beneficial to public health programming for primary cancer prevention in those regions.

Objectives

The aim of this project is to conduct background research on potential quantitative cancer prevention indicators, identify data sources, evaluate relevance and quality of indicators, link indicators to public health goals for cancer prevention, and propose a reporting framework specific to B.C.. This work was accomplished with the support of key informants from BC Cancer Health Promotion, the BCCDC and the BC Cancer Registry.

BC Cancer Health Promotion supports the health and wellness of British Columbians by increasing awareness, knowledge and adoption of cancer screening and prevention. Operations include:

1. Guidance for patients and providers regarding cancer prevention programs and services
2. External promotion to target populations
3. Community and stakeholder engagement
4. Knowledge mobilization and advocacy

Similar to reports by Cancer Care Ontario, the Health Promotion team within the Prevention, Screening and Hereditary Cancer Program at BC Cancer intends to generate an annual report consisting of quantitative indicators on the policies and programs addressing cancer risk factors and exposures in B.C. The findings in this document will help form this new initiative being undertaken.

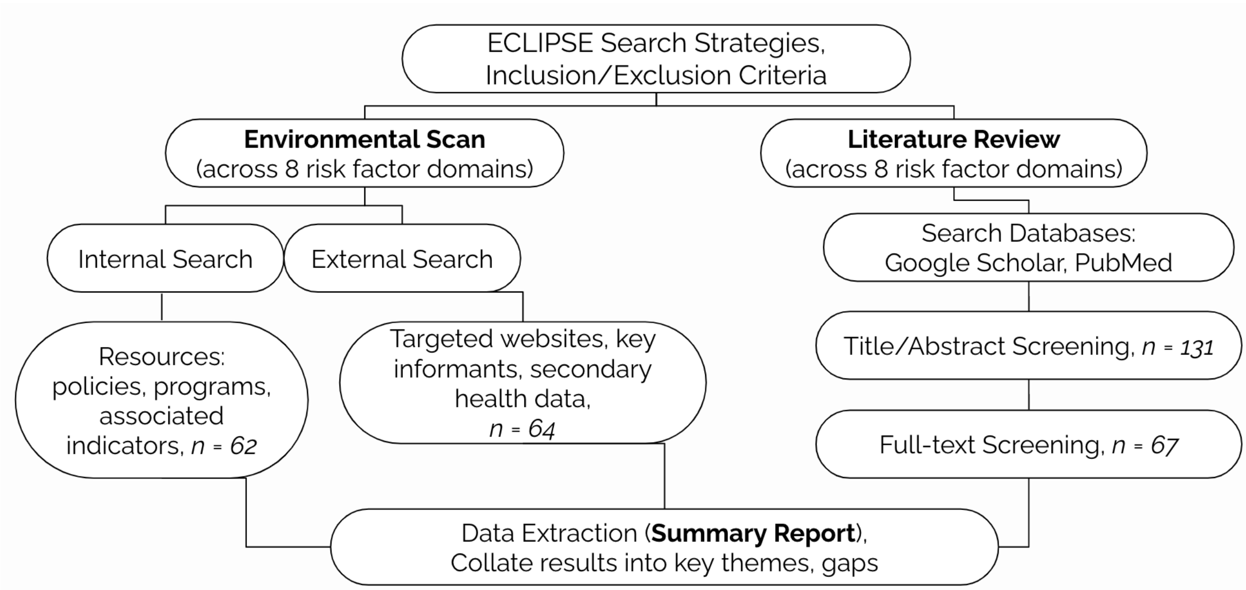
The following research questions utilized were:

1. What are the public health programs and policies aimed at cancer risk factor X in Canada and B.C.?
2. What is the prevalence of cancer risk factor X?
3. Are current programs and policies addressing cancer risk factor X effective as evidenced through evaluations?

Methods

Figure 1

Methods



To answer the above research questions, an environmental scan and literature review were conducted for all eight risk factor domains (smoking, alcohol, physical activity, diet, UV radiation, environment exposures, occupational exposures and infections). Search strategies and inclusion criteria were first formulated using the ECLIPSE (Expectation, Client Group, Location, Impact, Professionals, Service) framework (see Appendix B)[7]. Inclusion criteria: primary cancer prevention only, most recent policies and practices in B.C. within the last 10 years for all domains but occupational exposures (22 years), published resources accessible online, published in English.

For the environmental scan, internal resources were searched on the BC Cancer website including resources, activities and health promotion materials. External searches of gray literature were conducted using customized Google search engines, targeted websites and consultation with contact experts (see Appendix C). Across eight risk factor domains, 62 relevant resources were identified internally and 64 relevant resources externally (see Fig. 1). Resource types were subdivided into the following categories: cancer burden reports, cancer risk indicator reports, guiding principle reports for cancer control, health system performance reports and secondary health data.

To conduct the literature review, using the search strategies, keywords and terms were combined using boolean operators in Google Scholar and PubMed databases to retrieve relevant results (see Appendix B). Title/Abstract screening was conducted for 131 resources and full-text screening for 67 resources (see Fig. 1). Data from relevant resources were extracted and results collated into key themes and a gaps determination.

Data extracted from resources that met inclusion criteria within the environmental scan and literature review were subdivided into the following categories for further analysis: risk specific prevalence and cancer burden, economic costs, targeted policies and programs, indicators of policy and program implementation, evaluation methods and results and existing gaps and limitations[6].

Key Findings

SMOKING

Prevalence

Compared to Canada (10.3%), smoking rates in B.C. were 7.7% among adults age 15 and older in 2020, which was the lowest compared to other provinces and territories[8]. Prevalence of former smokers amongst adults was reported as 25.5% in B.C. and 24.8% in Canada[8]. In regards to cancer burden, attributable cancer cases due to tobacco smoking was 29.4% in Canada and 27.7% in B.C.[9]. Findings by Sweeney-Magee et al. demonstrate that non-smoking guidelines had the highest primary prevention adherence (94.8%) compared to guidelines addressing other risk factors (e.g. nonsmoking, alcohol consumption, diet) for colorectal cancer prevention[10].

Policies

- **Tobacco Control**

Current tobacco control strategies aim to decrease frequency and negative impacts of tobacco use through harm reduction[11]. These include taxation on all tobacco products including cigarettes, heated tobacco products, cigars and loose tobacco[12]. As of July 1, 2022, 7% PST has been implemented in addition to the tobacco tax[12]. Smoke-free area policies in B.C. include strata bylaws for apartment buildings and non-smoking areas on university campuses. The University of British Columbia and Simon Fraser University are currently implementing smoke-free policies in designated areas only[13, 14]. It is important to account for attitudes and behaviours of students, faculty and staff in achieving policy adherence when designing such policies[15].

- **Behavioural Interventions**

Most smoking cessation programs in B.C. focus on individual behavioural change. Such programs are intended to provide education and support in individual or group settings, and can consist of counseling, self-help resources and apps[16]. The most common tobacco cessation methods are quitting without aid, reducing the number of cigarettes smoked and nicotine replacement therapies (NRT) such as gum, lozenges and patches. Existing programs in B.C. addressing smoking cessation include QuitNow, HealthLinkBC: Quit Smoking, and Walk or Run to Quit. The BC Smoking Cessation Program also provides subsidization of NRTs for all ages. In regards to vaping, B.C. has released a 10-Point Vaping Action Plan, which focuses on youth programming, and use of regulatory authority to restrict access and exposure to vape products[17]. Lung cancer risk reduction recommendations as outlined by the CCS and WCRF include not smoking or using any form of tobacco and avoiding exposure through tobacco smoke. Additionally, workplace electronic campaigns targeting tobacco smoking through knowledge mobilization campaigns, awareness strategies, social media and text messages, and self-directed online programs exist in other parts of Canada[16].

Potential Data Sources: Canadian Tobacco and Nicotine Survey (2020), Canadian Student Tobacco, Alcohol and Drugs Survey (2019), Canadian Community Health Survey, ComPARe Study, Health Canada Provincial Sales Data (2019)

ALCOHOL

Consumption Guidelines

In 2011, national low-risk drinking guidelines (LRDGs) were developed in Canada to inform drinkers of associated harms with alcohol and to encourage lower-risk patterns of consumption[18]. Variation between different types of alcoholic drinks has not been shown to affect risk, and lower alcohol intake can reduce cancer risk associated with alcohol consumption[19]. The following measures have the most effect at reducing negative effects of alcohol consumption including cancer: increasing alcohol pricing and taxation, reductions in physical availability, restrictions on advertising and impaired driving countermeasures[20]. While zero consumption is considered the safest approach to alcohol-related cancer prevention, shifting to an abstinence approach is not feasible in Canada[21]. Cancer prevention organizations endorse Canada's LDRGs and the WCRF advises that it is best not to drink at all for cancer prevention. The CCS recommends less than one drink per day for women and less than two drinks per day for men for those who choose to drink alcohol[19]. Sweeney-Magee et al. found that individuals were significantly more adherent to alcohol recommendations (86.6%) compared to dietary (66.7%) or healthy body mass index (BMI) guidelines (44.2%)[10].

Prevalence

According to the Canadian Alcohol and Drugs Survey, past 12 month alcohol use was 77.2% in B.C. compared to 76.5% in Canada, while the percentage of British Columbians exceeding LDRGs was 17.1 compared to 17.6 in Canada[22]. More men (18%) than women (7%) drank above weekly guidelines[23]. Cancer was the most common cause of alcohol-contributable death in 2014 (31%)[23]. Attributable cancer cases due to alcohol in 2015 was 5.3% in B.C. compared to 5.1% in Canada[9]. Interestingly, 22% of Canadians reported a decrease in alcohol consumption since the start of the COVID-19 pandemic[24]. 33% of Canadians aged 15 to 29 and 18% aged 30 to 64 who reported having consumed alcohol in the previous month decreased their consumption during the pandemic[24].

Policies

- **Harm Reduction Strategies**

In Canada, distribution and sales of alcohol are controlled by a partial government monopoly[20]. Alcohol Policy and Cancer in Canada (2021) outlines B.C. regulations on minimum pricing, taxation and physical availability of alcoholic drinks[25]. Harm reduction strategies in B.C. include the Municipal Alcohol Policy (MAP) program, which is a policy tool aligning with provincial liquor laws and demonstrates careful alcohol use on public property such as parks, beaches, arenas, and community centres[26]. Alcohol labels with cancer warnings and national drinking guidelines may better convey risk information and promote safer consumption than other existing practices according to a study by Hobin et al.[27]. There has been moderate support in Canada for alcohol warning labels with health warnings, standard drink information and drinking guideline labels amongst survey respondents[28].

Potential Data Sources: WCRF 3rd Expert Report, ComPARe Study, BCCDC, CTNS (2020), Canadian Student Tobacco, Alcohol and Drugs Survey (2019), Alcohol Policy in Canada (2021)

PHYSICAL ACTIVITY

Exercise Guidelines

Prevention Guidelines for physical activity are fairly consistent amongst the WCRF, WHO and Canadian Society for Exercise Physiology[29]. In general, the recommendation is that one should be physically active, walk more and sit less. One should be at least moderately physically active by moving throughout the day by following or exceeding national guidelines, while limiting sedentary habits[10]. For Aerobic exercise, it is recommended that one performs at least 150 minutes of moderate intensity exercise per week or 75-150 mins of vigorous intensity exercise per week[29]. The WCRF and CCS recommend keeping weight as low as one can within the healthy BMI range throughout life (BMI = 18.5-24.9kg/m^2).

Prevalence

The World Health Organization (WHO) reports approximately 35% of cancer-related mortality is attributable to modifiable risk factors including lack of physical activity and obesity[30]. Physical activity rates in B.C. have been captured through PartipACTION report cards, which summarizes the current literature and national survey data, as well as the Canadian Community Health Survey[31]. Approximately 64% of adults age 18 and older in B.C. self-reported at least 150 minutes of physical activity per week[31]. Prevalence of moderate recreational activity in B.C. is 26.5% and low recreational physical activity is 40.9%[32]. B.C. has been shown to have the highest rates of moderate physical activity and lowest rates of inactive individuals amongst all provinces and territories[32]. In regards to sedentary behaviour, individuals who spend less than four hours a week exercising and more than five hours a week on screen based activities had a higher BMI than those who spend less than five hours on screen based activities[33]. Results of the ComPARE study show that attributable cancer cases due to physical inactivity is 8.4% in B.C., while sedentary behaviour is slightly higher in B.C. compared to the Canadian average, at 6.5% versus 5.7%[9]. According to an analysis of the Canadian Community Health Survey (2003) by Friedenreich et al., parameters for physical activity are: physically inactive = < 1.5 kcal/kg, moderately active = 1.5-2.9 kcal/kg, physical active >= 3.0 kcal/day[34].

Policies

Exercise Enabling Programs/Policies

Programs and policies in B.C. include the B.C. Physical Activity Strategy, which provides a framework for enhancing participation in physical activity in the province through collaboration with various stakeholders and local health authorities[35]. Additionally, resources that incorporate the built environment into provision of physical activity are available such as The BC Cycling Policy, Transcanada Trail BC and BCCDC's Healthy Built Environment Linkages Toolkit[56].

Potential Data Sources: Canadian Community Health Survey, ComPARe Study, ParticpACTION Report Cards, WCRF 3rd Expert Report

DIET

Dietary Guidelines

Dietary recommendations for cancer prevention by the WCRF are at least five servings per day of fruits and/or vegetables (1 serving = 125 ml of whole fruit, vegetable or vegetable juice, or 250 ml of leafy vegetables), and at least 30 g per day of dietary fiber, among others. Sweeney-Magee et al. found that in evaluating cancer prevention guidelines (CCS, WCRF) within a B.C. cohort, participants were less likely to adhere to fruit and vegetable consumption guidelines (66.7%) compared to the most adhered guidelines being nonsmoking (94.8%), physical activity (88.9%), and alcohol consumption (86.6%)[10].

Prevalence

The ComPARe study showcases the following factors attributable to cancer cases in B.C. to be: low fruit and vegetables, processed and red meat, and excess weight as defined by BMI (see Appendix D)[9]. Encouraged foods for cancer prevention include those high in nutrients and those that help one stay at a healthy body weight[36]. This includes incorporating a variety of vegetables (dark green, red, orange), fiber-rich legumes, fruits, and whole grains[36]. Additional foods that have been linked to increased cancer risk are sugar-sweetened beverages, highly processed foods and refined grain products[36].

Policies

- **Nutrition Programming**

Examples of nutrition programming in B.C. include free access to registered dieticians by HealthLinkBC, healthy eating programming for children and families by Live5210, and childhood obesity programming by ShapedownBC. Canada's healthy eating strategy (the food guide, how to read food labels, portion sizing) has shaped B.C.'s Guidelines for Food and Beverage Sales in Schools (2013) and Healthier Choices in Vending Machines (2014).

- **Social Determinants of Health**

Barriers to nutrition for cancer prevention include lack of food skills and safety, low socioeconomic status, and lack of access to healthy foods due to food deserts and other environmental factors[37]. The BCCDC has released resources focused on social determinants of health, such as Food Skills for Families[38] and the Food Security Indicator Framework[39]. Additionally, subsidization of local produce is available for underserved individuals and families through the BC Farmers Market Nutrition Program. Lack of affordable housing in less desirable neighborhoods has been shown to limit access to greenspace, healthy food stores and public services that contribute to maintaining a healthy diet[40].

A note on body weight

Brenner et al. estimate that at least 3.5% of all cancer cases are attributed to excess body weight, and the burden of disease is greater among women compared to men[41]. Statistics Canada (2018) reports 23.1% of the adult population in B.C. are obese, compared to the national average of 26.8%. While weight has not been formally included as a risk factor domain in this report, it is important to highlight that many indicators, programs and policies addressing diet and physical activity have overlapping goals for maintaining a healthy weight.

Potential Data Sources: WCRF 3rd Expert Report on Cancer Prevention and Survival (2018), Canadian Community Health Survey, BCCDC Indicator Library, Household Food Insecurity Indicator Report (2016)

ULTRAVIOLET RADIATION

Sun Exposure Guidelines

Primary prevention guidelines as outlined by the WCRF consists of reducing UV exposure by limiting time in the sun by checking the UV index provided by Environment Canada, seeking shade, wearing protective clothing, using broad-spectrum sunscreens and avoiding tanning beds[42,43]. Sun exposure guidelines in B.C. are administered by HealthLinkBC, Sun Safe BC (BC Cancer) and SunSense (CCS).

Prevalence

The largest contributing risk factor for all skin cancers is exposure to UV radiation by the sun and tanning beds[44]. According to the ComPARe study (2015), attributable cancer cases due to UV risk behaviours was 54% in B.C.[9]. All skin cancers are more common in fair-skinned individuals[44]. Skin type is often categorized using the Fitzpatrick Skin-Typing Scale, which categorizes skin types from I-V by skin colour and ease of burn, with I and II being at highest risk of skin cancer[45]. UV exposure risk is compounded by outdoor activities such as golfing, hiking and swimming, and for those that vacation in warmer climates[46]. An analysis of the Canadian Community Health Survey found that 17% of Canadians spent four or more hours in the sun and 33% reported having had a sunburn in the past year, with men more likely to report a sunburn than women[47]. Skin damage due to UV radiation accumulates over time, with 23% of photodamage occurring by age 18, 46% by age 40 and 74% by age 59[48].

Policies

- **Sun Safety Programming**

Five priority areas for sun safety implementation include: child care centers, elementary schools, outdoor occupational settings, outdoor recreational settings and multicomponent community-wide interventions[42]. Unlike other countries such as Australia, sun safety in Canada is not centralized, including programs and funding sources. It is a collaborative effort by researchers and organizations including the Canadian Dermatology Association (CDA), CCS, Health Canada, provincial and territorial ministries of health and local public health units[49]. In B.C., Sun Safe BC facilitates a coalition of contact experts working to reduce exposure through policy and health promotion.

- **Children and Youth**

Skin cancer control is a priority especially among young children, as risks for melanoma arise as early as infancy, and can accumulate into adulthood easier without interventions targeting environments occupied by children such as daycares and schools[43]. Sun Safe BC conducts health promotion projects such as online training for early childhood educators, shade pilot studies at elementary schools and sunscreen dispenser programming[50]. As of October 2012, tanning bed usage by minors under 18 years old is banned in B.C. unless a medical prescription has been issued[51].

Possible Data Sources: Canadian cancer statistics, ComPARe Study, Canadian Community Health Survey, Canadian Dermatology Association, WCRF, CCS

ENVIRONMENTAL EXPOSURES

Prevalence

Environmental indicators for carcinogens prioritized by CAREX Canada broadly consist of outdoor air, indoor air, indoor dust, drinking water, foods, and beverages[52]. Environmental factors account for 4.1% (7636 cases) of cancers in Canada including UV radiation, outdoor air pollution, and residential radon[2].

Air pollution is the most commonly measured outdoor air pollutant in Canada, and is composed of particulate matter (PM), ground-level ozone, carbon monoxide, sulfur dioxide, and nitrogen oxides, which constitute smog and acid rain[1]. Attributable cancer cases due to outdoor air pollution was 6% in B.C. compared to 6.9% in Canada. Prevalence data can also be found in the Air Quality Health Index for B.C.. Additionally, wildfires may emit carcinogens that contaminate outdoor and indoor environments, but little is known about the relationship between exposure to wildfires and cancer risk at present. A population-based observational cohort study by Korsiak et al. suggests that wildfire exposure is associated with slightly increased incidence of lung cancer and brain tumors[53].

The ComPARe study (2015) showcases attributable cancer cases due to residential radon as 4.4% in B.C. compared to the Canadian average (6.9%)[9]. It is predicted that if all residential radon exposures above Canadian policy guidelines (200 Mg/m^3) were reduced to 50 Bg/m^3, 293 cancer cases could be prevented by 2042, and 2322 cumulative cases could be prevented between 2016 and 2042 in Canada[54]. Prevalence data for residential radon is provided by the BCCDC Radon Map which includes indoor radon gas exposure levels and number of homes tested that were above minimum threshold.

Policies

In 2007, recommendations from the National Committee on Environmental and Occupational Exposures, the CCS and the Canadian Partnership Against Cancer led to initiation of the CARcinogen EXposure (CAREX) Canada project for primary prevention[52]. The goal of CAREX Canada is to develop and implement exposure surveillance methods for known or suspected carcinogens, and is composed of an occupational arm and environmental arm[52].

- **Air Quality Control**

Air quality and PM goals have been outlined in PM_{10} and $PM_{2.5}$ Provincial Objectives and Standards and the CCME Guidance Document on Achievement Determination on Canadian Ambient Air Quality Standards for Fine PM and Ozone. HealthLinkBC and the BCCDC provide descriptions of common air pollutants, and instructions for reducing exposure to PM for homeowners. The BCCDC also provides smoke reduction strategies, vehicle emission strategies and wildfire smoke fact sheets.

- **Radon Awareness and Testing**

The WHO (2009) recommends increasing ventilation indoors where radon can accumulate, reducing negative pressures within buildings to prevent radon inflow from the ground and establishing national awareness and mitigation programs[55]. While there is no safe concentration for radon gas, guidelines provide threshold concentrations to map areas of higher risk. Policy highlights include radon testing and mitigation protocols by BCCDC and awareness programs such as the BC Lung Association: RadonAware.

- **Built Environment**

In addressing the built environment, the Healthy Built Environment Linkages Toolkit outlines how neighbourhood design, transportation networks, natural environments, food systems and housing influence outcomes related to social well-being, economic co-benefits and communities of various sizes in B.C.[56]. The toolkit also discusses community planning and design recommendations for reducing environmental hazard exposure[56].

Potential Data Sources: Health Canada: Cross-Canada Survey on Radon Concentrations in Homes (2012), CAREX Canada: Resources on environmental and occupational exposures and cancer risk, Air Quality Health Index (AQHI) for BC, PM10 and PM2.5 Provincial Objectives and Standards, CCME Guidance Document on Achievement Determinants on Canadian Ambient Air, Canadian Partnership Against Cancer

OCCUPATIONAL EXPOSURES

Prevalence

CAREX Canada has prioritized 44 carcinogens, with level of exposure estimates for 18 agents [57]. The top 10 occupational exposures in B.C. are solar radiation, shift work, gasoline/diesel engine exhaust, polycyclic aromatic hydrocarbons, silica, wood dust, secondhand smoke, benzene, welding fumes and asbestos (see Appendix E). In 2011, approximately 3.9% (7700 cases) - 4.2% (21,800 cases) of all incident cancers were due to occupational exposure, where the top cancer sites were: mesothelioma, non-melanoma skin cancer, lung, breast, and bladder[58]. Most workers exposed were male, and industries associated with the highest occupational exposure risk for cancer were agriculture, construction, mining, transport, trades, and equipment operation [59, 60, 61].

CAREX Canada has identified UV radiation as the second most prominent carcinogenic exposure in Canada, where over 75% of outdoor workers operate within the highest exposure category[62]. Further, diesel engine exhaust is the most prevalent occupational lung carcinogen in Canada, where the majority of cancer burden is attributable to low concentrations[60]. It is estimated that the number of workers exposed and level of exposure to second hand smoke in Canada decreased by 20% between 2006-2016[61]. Additionally, it has been shown that the greatest cancer burden attributable to radon occurs at low levels of occupational exposure[63]. Approximately 15.5 million Canadian employees were exposed to radon during the risk exposure period of 1961-2001 according to the Canadian Census and Labour Force Survey[63]. Polycyclic Aromatic Hydrocarbons are a group of environmental pollutants that have been known to contain carcinogens associated with breast and other cancer sites[64].

Policies

- **Reduction/Elimination Strategies**

In order of effectiveness for controlling risk, strategies to protect workers from exposures to harmful carcinogens as outlined by the Canadian Centre for Occupational Health and Safety (2018) are: Elimination, Engineering controls, Administrative controls, Personal protective equipment (PPE)[65].

- **Awareness Programming**

Awareness program highlights include Sun Safety at Work which supports workplaces to assess exposure risks, implement control strategies for existing programs and integrate controls into existing occupational health and safety systems[62]. Health promotion and public health messaging efforts specialized for outdoor workers have been generated as part of the Outdoor Workers Project in Vancouver, B.C. which provides evidence-based sun avoidance recommendations including peak UVR times, shady breaks, increased sun protection and task reorganization[66].

- **Priority Setting**

CAREX Canada (2015) has established priorities for preventing occupational cancer using four criteria[67]:

1. Likelihood of presence and/or use in Canadian workplaces
2. Toxicity of substance (carcinogenicity and other health effects)
3. Feasibility of producing a carcinogen profile and/or an occupational estimate

4. Special interest from the public and stakeholders

Potential Data Sources: Canadian Partnership Against Cancer, Canadian Workplace Exposure Database, CAREX Occupational Exposure to Radon and Other Hazard Statistics

INFECTIONS

Prevalence

Infections (HPV, hepatitis B and others) contributed 3.7% of all cancer burden among adults in Canada in 2015[68]. HPV causes 99% of the cervical cancer burden in Canada, and approximately 200 cases are reported annually in B.C.[69]. According to Canadian Notifiable Disease Surveillance system data, reported rates for chronic hepatitis B in B.C. is 14.7 per 100,000 individuals[70]. Liver cancer burden due to hepatitis B is 8.8% in Canada[9]. In 2018, 460 new liver cancer cases were diagnosed in B.C.[71]. Immunization data in B.C. is captured by the BCCDC Reportable Diseases Dashboard which contains summary statistics and data visualizations sorted by health region. Additionally, BC Immunization coverage reports include data for each health authority up to 2020. HPV vaccination coverage rates in B.C. is 66.1% for grade six female children (52%-78% range) and 63.5% for grade six male children (48%-74% range)[72].

Policies

- **Immunization Programming**

In Canada, opportunistic cervical cancer prevention approaches occur at the provincial level, with HPV vaccination being the main approach for primary prevention[73]. The two prophylactic HPV vaccines utilized in B.C. are Gardasil (HPV9), which is approved for use in both males and females, and Cervarix (HPV2)[74]. HPV vaccination programs are primarily school-based and publicly funded for the target age group, resulting in higher coverage[73]. As of October 2016, all 13 Canadian jurisdictions vaccinate girls and six jurisdictions including B.C. also vaccinate boys in school-based publicly funded programs[75]. HPV immunization programs in B.C. consist of a routine schedule (0, 2, 6 months) beginning in grade six, and catch up programs in grade nine[73].

Primary prevention of hepatitis B virus infection is accomplished by universal vaccination in infancy[76]. Hepatitis B vaccine is provided in B.C. for free to infants as part of a routine vaccination schedule, with catch up programs for children in grade six[77]. It is also provided free to individuals at high risk for infection including children whose familes have emigrated from high-risk regions, individuals who may have sexual contact with someone with hepatitis B, those with chronic liver and/or kidney disease and others[77].

- **Health Promotion**

HealthLinkBC produces online resources for the public regarding HPV and hepatitis B vaccines. Health promotion strategies for cancer prevention including informational pamphlets and resources for how to access immunizations have been used by HealthLinkBC and ImmunizeBC. Additionally, Kids Boost Immunity is a collaboration by the Public Health Association of BC and UNICEF, containing a series of over 200 free curriculum-linked lessons for teachers[78]. Kids Boost offers programs for grades 4-12[78].

Potential Data Sources: BC Immunization Coverage Reports, BCCDC Indicator Library, BCCDC Reportable Diseases Dashboard, Childhood Immunization Coverage Dashboard

Conclusion

Interventions targeting modifiable cancer risk factors may contribute to the prevention of 10,600 to 39,700 cases in Canada by 2042[2]. Findings from the environmental scan and literature review will require further synthesis as priority indicators are identified by BC Cancer. This work highlights the importance of prioritizing the development and implementation of cancer prevention strategies at the individual, community and societal levels.

Opportunities for cancer prevention activities that address multiple risk factor domains should be considered through operationalization of the built environment and community-based programs and policies. Within each risk factor domain, priority indicators should be further stratified by geographical residence, household income, education, neighbourhood, and other socioeconomic factors, as social determinants of health can highlight disparities impacting indicators. Additionally, reportable disease data may not be up to date due to COVID-19 priorities. Further research is required to evaluate linkages between vaping and cancer incidence. Primary prevention program resiliency in regards to COVID-19, and actions specific to Indigenous health should also be areas of interest for future work.

References

1. Pader, J., Ruan, Y., Poirier, A. E., Asakawa, K., Lu, C., Memon, S., ... & Brenner, D. R. (2021). Estimates of future cancer mortality attributable to modifiable risk factors in Canada. *Canadian Journal of Public Health*, *112*(6), 1069-1082.
2. Poirier, A. E., Ruan, Y., Volesky, K. D., King, W. D., O'Sullivan, D. E., Gogna, P., Walter, S. D., Villeneuve, P. J., Friedenreich, C. M., Brenner, D. R., & ComPARe Study Team (2019). The current and future burden of cancer attributable to modifiable risk factors in Canada: Summary of results. *Preventive medicine*, *122*, 140–147. https://doi.org/10.1016/j.ypmed.2019.04.007
3. Lee, S. (n.d.). *What causes cancer?* Canadian Cancer Society. Retrieved July 27, 2022, from https://cancer.ca/en/cancer-information/what-is-cancer/what-causes-cancer
4. *Cancer risk factors that affect cancer: Exposures:* WCRF International. (2022, April 21). Retrieved July 27, 2022, from https://www.wcrf.org/diet-activity-and-cancer/risk-factors/
5. Brenner, D. R., Poirier, A., Woods, R. R., Ellison, L. F., Billette, J. M., Demers, A. A., Zhang, S. X., Yao, C., Finley, C., Fitzgerald, N., Saint-Jacques, N., Shack, L., Turner, D., Holmes, E., & Canadian Cancer Statistics Advisory Committee (2022). Projected estimates of cancer in Canada in 2022. *CMAJ : Canadian Medical Association journal = journal de l'Association médicale canadienne*, *194*(17), E601–E607. https://doi.org/10.1503/cmaj.212097
6. Ontario Health (Cancer Care Ontario). Prevention System Quality Index 2020. Toronto: Queen's Printer for Ontario; 2020.
7. Wildridge, V., & Bell, L. (2002). How CLIP became ECLIPSE: a mnemonic to assist in searching for health policy/management information. *Health Information & Libraries Journal*, *19*(2), 113-115. World Cancer Research Fund/American Institute for Cancer Research. *Diet, Nutrition, Physical Activity and Cancer: a Global Perspective.* Continuous Update Project Expert Report 2018. Available at dietandcancerreport.org
8. Health Canada. (2022, April 1). *Canadian Tobacco and Nicotine Survey (CTNS): summary of results for 2020*. Canada.ca. Retrieved July 27, 2022, from https://www.canada.ca/en/health-canada/services/canadian-tobacco-nicotine-survey/2020-summary.html
9. ComPARe. (2015). *Compare study on preventable cancers in Canada*. Retrieved July 27, 2022, from https://prevent.cancer.ca/
10. Sweeney-Magee, M., Gotay, C., Karim, M. E., Telford, J., & Dummer, T. (2022). Patterns and determinants of adherence to colorectal cancer primary and secondary prevention recommendations in the BC Generations Project. Tendances et déterminants de l'adhésion aux recommandations en matière de prévention primaire et secondaire du cancer colorectal dans le cadre du BC Generations Project. *Health promotion and chronic disease prevention in Canada : research, policy and practice*, *42*(2), 79–93. https://doi.org/10.24095/hpcdp.42.2.04
11. Singhavi, H., Ahluwalia, J. S., Stepanov, I., Gupta, P. C., Gota, V., Chaturvedi, P., & Khariwala, S. S. (2018). Tobacco carcinogen research to aid understanding of cancer risk and influence policy. *Laryngoscope investigative otolaryngology*, *3*(5), 372-376.

12. Ministry of Finance. (2022, July 15). *Tobacco tax*. Province of British Columbia. Retrieved July 27, 2022, fromhttps://www2.gov.bc.ca/gov/content/taxes/sales-taxes/tobacco-tax#:~:text=Effective%20July%201%2C%202022%2C%20PST,Tobacco%20Sellers%20for%20more%20information.
13. The University of British Columbia. (n.d.). *Smoke-and-Vape-Free Areas*. Smoke-and-vape-free areas. Retrieved July 27, 2022, from https://wellbeing.ubc.ca/smokefree
14. *Smoking on campus*. Smoking - Work & research safety - Simon Fraser University. (n.d.). Retrieved July 27, 2022, from https://www.sfu.ca/srs/work-research-safety/general-safety/smoking.html
15. Dilliott, D., Fazel, S., Ehsan, N., & Sibbald, S. L. (2020). The attitudes and behaviors of students, staff and faculty towards smoke-free and tobacco-free campus policies in North American universities: A narrative review. *Tobacco prevention & cessation*, *6*, 47. https://doi.org/10.18332/tpc/125080
16. Hui, D., & Mierzwinski-Urban, M. (2021). *Workplace Electronic Health Promotion Campaigns for Tobacco Smoking Prevention or Cessation*. Canadian Agency for Drugs and Technologies in Health.
17. *Reducing Youth Access & Exposure to Vapour Products: A PlanH Resource*. Planh. (1970, January Retrieved July 27, 2022, from
https://planh.ca/resources/action-guides/reducing-youth-access-exposure-vapour-products-planh-resource
18. Vallance, K., Romanovska, I., Stockwell, T., Hammond, D., Rosella, L., & Hobin, E. (2018). "We Have a Right to Know": Exploring Consumer Opinions on Content, Design and Acceptability of Enhanced Alcohol Labels. *Alcohol and alcoholism (Oxford, Oxfordshire)*, *53*(1), 20–25. https://doi.org/10.1093/alcalc/agx068
19. Amin, G., Siegel, M., & Naimi, T. (2018). National Cancer Societies and their public statements on alcohol consumption and cancer risk. *Addiction (Abingdon, England)*, *113*(10), 1802–1808. https://doi.org/10.1111/add.14254
20. Vallance, K., Vincent, A., Schoueri-Mychasiw, N., Stockwell, T., Hammond, D., Greenfield, T. K., McGavock, J., & Hobin, E. (2020). News Media and the Influence of the Alcohol Industry: An Analysis of Media Coverage of Alcohol Warning Labels With a Cancer Message in Canada and Ireland. *Journal of studies on alcohol and drugs*, *81*(2), 273–283. https://doi.org/10.15288/jsad.2020.81.273
21. Grevers, X., Ruan, Y., Poirier, A. E., Walter, S. D., Villeneuve, P. J., Friedenreich, C. M., Brenner, D. R., & ComPARe Study Team (2019). Estimates of the current and future burden of cancer attributable to alcohol consumption in Canada. *Preventive medicine*, *122*, 40–48. https://doi.org/10.1016/j.ypmed.2019.03.020
22. Health Canada. (2021, December 20). *Canadian Alcohol and Drugs Survey (CADS): summary of results for 2019*. Canada.ca. Retrieved July 27, 2022, from
https://www.canada.ca/en/health-canada/services/canadian-alcohol-drugs-survey/2019-summary.html
23. Sherk, A., Thomas, G., Churchill, S., & Stockwell, T. (2020). Does Drinking Within Low-Risk Guidelines Prevent Harm? Implications for High-Income Countries Using the International Model of Alcohol Harms and Policies. *Journal of studies on alcohol and drugs*, *81*(3), 352–361.

24. Government of Canada (2021, March 4). *Alcohol and cannabis use during the pandemic: Canadian Perspectives Survey Series 6*. The Daily - . Retrieved July 28, 2022, from https://www150.statcan.gc.ca/n1/daily-quotidien/210304/dq210304a-eng.htm
25. Canadian Partnership Against Cancer. (2021, December 13). *Alcohol policy and cancer in Canada*. Canadian Partnership Against Cancer. Retrieved July 27, 2022, from https://www.partnershipagainstcancer.ca/topics/alcohol-policies/
26. *The Municipal Alcohol Policy Program in British Columbia - ccsa.ca*. (n.d.). Retrieved July 27, 2022, From
https://www.ccsa.ca/sites/default/files/2019-04/CCSA-Municipal-Alcohol-Policy-British-Columbia-2017-en_0.pdf
27. Hobin, E., Shokar, S., Vallance, K., Hammond, D., McGavock, J., Greenfield, T. K., Schoueri-Mychasiw, N., Paradis, C., & Stockwell, T. (2020). Communicating risks to drinkers: testing alcohol labels with a cancer warning and national drinking guidelines in Canada. *Canadian journal of public health = Revue canadienne de sante publique*, *111*(5), 716–725. https://doi.org/10.17269/s41997-020-00320-7
28. Vallance, K., Stockwell, T., Zhao, J., Shokar, S., Schoueri-Mychasiw, N., Hammond, D., Greenfield, T. K., McGavock, J., Weerasinghe, A., & Hobin, E. (2020). Baseline Assessment of Alcohol-Related Knowledge of and Support for Alcohol Warning Labels Among Alcohol Consumers in Northern Canada and Associations With Key Sociodemographic Characteristics. *Journal of studies on alcohol and drugs*, *81*(2), 238–248. https://doi.org/10.15288/jsad.2020.81.238
29. Bandera, E. V., Alfano, C. M., Qin, B., Kang, D. W., Friel, C. P., & Dieli-Conwright, C. M. (2021). Harnessing nutrition and physical activity for breast cancer prevention and control to reduce racial/ethnic cancer health disparities. *American Society of Clinical Oncology Educational Book*, *41*, e62-e78.
30. Lugo, D., Pulido, A. L., Mihos, C. G., Issa, O., Cusnir, M., Horvath, S. A., Lin, J., & Santana, O. (2019). The effects of physical activity on cancer prevention, treatment and prognosis: A review of the literature. *Complementary therapies in medicine*, *44*, 9–13. https://doi.org/10.1016/j.ctim.2019.03.013
31. Government of Canada. (2022, April 13). *Canadian Community Health Survey - annual component (CCHS)*. Surveys and statistical programs. Retrieved July 27, 2022, from https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=3226
32. Government of Canada. (2007, October 24). *Canadian Community Health Survey (CCHS)*. Surveys and statistical programs. Retrieved July 27, 2022, from https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&Id=4995
33. Chow R. (2017). Decreasing screen time and/or increasing exercise only helps in certain situations for young adults. *International journal of adolescent medicine and health*, *32*(2), /j/ijamh.2020.32.issue-2/ijamh-2017-0100/ijamh-2017-0100.xml. https://doi.org/10.1515/ijamh-2017-0100
34. Friedenreich, C. M., Barberio, A. M., Pader, J., Poirier, A. E., Ruan, Y., Grevers, X., Walter, S. D., Villeneuve, P. J., Brenner, D. R., & ComPARe Study Team (2019). Estimates of the current and future burden of cancer attributable to lack of physical activity in Canada. *Preventive medicine*, *122*, 65–72. https://doi.org/10.1016/j.ypmed.2019.03.008

35. Ministry of Health. (2015, November). *Active people, active places - BC physical activity strategy*. Retrieved July 27, 2022, from https://www.health.gov.bc.ca/library/publications/year/2015/active-people-active-places-web-2015.pdf
36. Murphy, R. A. (2022). Diet, Physical Activity, and Cancer Prevention. In *Nutrition Guide for Physicians and Related Healthcare Professions* (pp. 149-158). Humana, Cham.x
37. Di Sebastiano, K. M., Murthy, G., Campbell, K. L., Desroches, S., & Murphy, R. A. (2019). Nutrition and Cancer Prevention: Why is the Evidence Lost in Translation?. *Advances in nutrition (Bethesda, Md.)*, *10*(3), 410–418. https://doi.org/10.1093/advances/nmy089
38. *Food skills for families*. BC Centre for Disease Control. (n.d.). Retrieved July 28, 2022, from http://www.bccdc.ca/our-services/programs/food-skills-for-families
39. Seed, B., Kurrein, M., & Hasdell, R. (2022). A Food Security Indicator Framework for British Columbia, Canada. *Health promotion practice*, 15248399211073801. Advance online publication. https://doi.org/10.1177/15248399211073801
40. Wray, A., & Minaker, L. M. (2019). Is cancer prevention influenced by the built environment? A multidisciplinary scoping review. *Cancer*, *125*(19), 3299–3311. https://doi.org/10.1002/cncr.32376
41. Brenner D. R. (2014). Cancer incidence due to excess body weight and leisure-time physical inactivity in Canada: implications for prevention. *Preventive medicine*, *66*, 131–139. https://doi.org/10.1016/j.ypmed.2014.06.018
42. LoConte, N. K., Gershenwald, J. E., Thomson, C. A., Crane, T. E., Harmon, G. E., & Rechis, R. (2018). Lifestyle modifications and policy implications for primary and secondary cancer prevention: diet, exercise, sun safety, and alcohol reduction. *American Society of Clinical Oncology Educational Book*, *38*, 88-100.
43. Coffin, T., Wu, Y. P., Mays, D., Rini, C., Tercyak, K. P., & Bowen, D. (2019). Relationship of parent-child sun protection among those at risk for and surviving with melanoma: Implications for family-based cancer prevention. *Translational behavioral medicine*, *9*(3), 480–488. https://doi.org/10.1093/tbm/ibz032
44. Andrews, H. (2012). Skin and sun awareness and skin cancer prevention. *British Journal of Healthcare Assistants*, *6*(12), 582-588.
45. Ministry of Health. (2016, January 8). *Protecting yourself from ultraviolet (UV) radiation*. Province of British Columbia. Retrieved July 27, 2022, from https://www2.gov.bc.ca/gov/content/health/keeping-bc-healthy-safe/radiation/ultraviolet-uv-radiation/protecting-yourself-from-ultraviolet-uv-radiation
46. Buller, D. B., Andersen, P. A., Walkosz, B. J., Scott, M. D., Beck, L., & Cutter, G. R. (2016). Rationale, design, samples, and baseline sun protection in a randomized trial on a skin cancer prevention intervention in resort environments. *Contemporary clinical trials*, *46*, 67–76. https://doi.org/10.1016/j.cct.2015.11.015
47. Pinault, L., & Fioletov, V. (2017). Sun exposure, sun protection and sunburn among Canadian adults. *Health reports*, *28*(5), 12–19.
48. Barber, K., Searles, G. E., Vender, R., Teoh, H., Ashkenas, J., & Canadian Non-melanoma Skin Cancer Guidelines Committee (2015). Non-melanoma Skin Cancer in Canada Chapter 2: Primary

Prevention of Non-melanoma Skin Cancer. *Journal of cutaneous medicine and surgery*, *19*(3), 216–226. https://doi.org/10.1177/1203475415576465
49. Garvin, T., & Eyles, J. (2001). Public health responses for skin cancer prevention: the policy framing of Sun Safety in Australia, Canada and England. *Social science & medicine (1982)*, *53*(9), 1175–1189. https://doi.org/10.1016/s0277-9536(00)00418-4
50. Sun safe BC - save your skin foundation. Save Your Skin Foundation -. (2020, December 16). Retrieved July 27, 2022, from https://saveyourskin.ca/sun-safe-bc/
51. Ministry of Health. (2017, April 11). *B.C. Tanning Bed Ban*. Province of British Columbia. Retrieved July 27, 2022, from https://www2.gov.bc.ca/gov/content/health/keeping-bc-healthy-safe/pses-mpes/tanning-beds/bc-tanning-bed-ban
52. Setton, E., Hystad, P., Poplawski, K., Cheasley, R., Cervantes-Larios, A., Keller, C. P., & Demers, P. A. (2013). Risk-based indicators of Canadians' exposures to environmental carcinogens. *Environmental health : a global access science source*, *12*, 15. https://doi.org/10.1186/1476-069X-12-15
53. Korsiak, J., Pinault, L., Christidis, T., Burnett, R. T., Abrahamowicz, M., & Weichenthal, S. (2022). Long-term exposure to wildfires and cancer incidence in Canada: a population-based observational cohort study. *The Lancet. Planetary health*, *6*(5), e400–e409. https://doi.org/10.1016/S2542-5196(22)00067-5
54. Gogna, P., Narain, T. A., O'Sullivan, D. E., Villeneuve, P. J., Demers, P. A., Hystad, P., Brenner, D. R., Friedenreich, C. M., King, W. D., & ComPARe Study Team (2019). Estimates of the current and future burden of lung cancer attributable to residential radon exposure in Canada. *Preventive medicine*, *122*, 100–108. https://doi.org/10.1016/j.ypmed.2019.04.005
55. Espina, C., Porta, M., Schüz, J., Aguado, I. H., Percival, R. V., Dora, C., Slevin, T., Guzman, J. R., Meredith, T., Landrigan, P. J., & Neira, M. (2013). Environmental and occupational interventions for primary prevention of cancer: a cross-sectoral policy framework. *Environmental health perspectives*, *121*(4), 420–426. https://doi.org/10.1289/ehp.1205897
56. BC Centre for Disease Control. Healthy Built Environment Linkages Toolkit: making the links between design, planning and health, Version 2.0. Vancouver, B.C. Provincial Health Services Authority, 2018. from http://www.bccdc.ca/pop-public-health/Documents/HBE_linkages_toolkit_2018.pdf
57. Peters, C. E., Ge, C. B., Hall, A. L., Davies, H. W., & Demers, P. A. (2015). CAREX Canada: an enhanced model for assessing occupational carcinogen exposure. *Occupational and environmental medicine*, *72*(1), 64–71. https://doi.org/10.1136/oemed-2014-102286
58. Labrèche, F., Kim, J., Song, C., Pahwa, M., Ge, C. B., Arrandale, V. H., McLeod, C. B., Peters, C. E., Lavoué, J., Davies, H. W., Nicol, A. M., & Demers, P. A. (2019). The current burden of cancer attributable to occupational exposures in Canada. *Preventive medicine*, *122*, 128–139. https://doi.org/10.1016/j.ypmed.2019.03.016
59. Peters, C. E., Kim, J., Song, C., Heer, E., Arrandale, V. H., Pahwa, M., Labrèche, F., McLeod, C. B., Davies, H. W., Ge, C. B., & Demers, P. A. (2019). Burden of non-melanoma skin cancer attributable to occupational sun exposure in Canada. *International archives of occupational and environmental health*, *92*(8), 1151–1157. https://doi.org/10.1007/s00420-019-01454-z

60. Kim, J., Peters, C. E., Arrandale, V. H., Labrèche, F., Ge, C. B., McLeod, C. B., Song, C., Lavoué, J., Davies, H. W., Nicol, A. M., Pahwa, M., & Demers, P. A. (2018). Burden of lung cancer attributable to occupational diesel engine exhaust exposure in Canada. *Occupational and environmental medicine*, *75*(9), 617–622. https://doi.org/10.1136/oemed-2017-104950

61. Rydz, E., Arrandale, V. H., & Peters, C. E. (2020). Population-level estimates of workplace exposure to secondhand smoke in Canada. *Canadian journal of public health = Revue canadienne de sante publique*, *111*(1), 125–133. https://doi.org/10.17269/s41997-019-00252-x

62. Kramer, D. M., Tenkate, T., Strahlendorf, P., Kushner, R., Gardner, A., & Holness, D. L. (2015). Sun Safety at Work Canada: a multiple case-study protocol to develop sun safety and heat protection programs and policies for outdoor workers. *Implementation science : IS*, *10*, 97. https://doi.org/10.1186/s13012-015-0277-2

63. Ge, C. B., Kim, J., Labrèche, F., Heer, E., Song, C., Arrandale, V. H., Pahwa, M., Peters, C. E., & Demers, P. A. (2020). Estimating the burden of lung cancer in Canada attributed to occupational radon exposure using a novel exposure assessment method. *International archives of occupational and environmental health*, *93*(7), 871–876. https://doi.org/10.1007/s00420-020-01537-2

64. Lee, D. G., Burstyn, I., Lai, A. S., Grundy, A., Friesen, M. C., Aronson, K. J., & Spinelli, J. J. (2019). Women's occupational exposure to polycyclic aromatic hydrocarbons and risk of breast cancer. *Occupational and environmental medicine*, *76*(1), 22–29. https://doi.org/10.1136/oemed-2018-105261

65. Canadian Centre for Occupational Health and Safety. (2022, July 27). *Hazard control : Osh answers*. Canadian Centre for Occupational Health and Safety. Retrieved July 27, 2022, from https://www.ccohs.ca/oshanswers/hsprograms/hazard_control.html

66. Peters, C. E., Tenkate, T., Heer, E., O'Reilly, R., Kalia, S., & Koehoorn, M. W. (2020). Strategic Task and Break Timing to Reduce Ultraviolet Radiation Exposure in Outdoor Workers. *Frontiers in public health*, *8*, 354. https://doi.org/10.3389/fpubh.2020.00354

67. Peters, C. E., Palmer, A. L., Telfer, J., Ge, C. B., Hall, A. L., Davies, H. W., Pahwa, M., & Demers, P. A. (2018). Priority Setting for Occupational Cancer Prevention. *Safety and health at work*, *9*(2), 133–139. https://doi.org/10.1016/j.shaw.2017.07.005

68. Volesky, K. D., El-Zein, M., Franco, E. L., Brenner, D. R., Friedenreich, C. M., Ruan, Y., & ComPARe Study Team (2019). Cancers attributable to infections in Canada. *Preventive medicine*, *122*, 109–117. https://doi.org/10.1016/j.ypmed.2019.03.035

69. Statistics by cancer type - cervix. (n.d.). Retrieved July 27, 2022, from http://www.bccancer.bc.ca/statistics-and-reports-site/Documents/Cancer_Type_Cervix_2018_20210305.pdf

70. Public Health Agency of Canada. (2021, July 28). *Hepatitis B in Canada: 2019 surveillance data*. Canada.ca. Retrieved July 27, 2022, from https://www.canada.ca/en/public-health/services/publications/diseases-conditions/hepatitis-b-2019-surveillance-data.html

71. *Statistics by cancer type - liver*. (n.d.). Retrieved July 27, 2022, from http://www.bccancer.bc.ca/statistics-and-reports-site/Documents/Cancer_Type_Liver_2018_20210305.pdf

72. Childhood immunization coverage dashboard. BC Centre for Disease Control. (n.d.). Retrieved July 27, 2022, from http://www.bccdc.ca/health-professionals/data-reports/childhood-immunization-coverage-Dashboard

73. Saraiya, M., Steben, M., Watson, M., & Markowitz, L. (2013). Evolution of cervical cancer screening and prevention in United States and Canada: implications for public health practitioners and clinicians. *Preventive medicine*, *57*(5), 426–433. https://doi.org/10.1016/j.ypmed.2013.01.020

74. HPV. Immunize BC. (2022, July 22). Retrieved July 27, 2022, from https://immunizebc.ca/hpv

75. Shapiro, G. K., Guichon, J., & Kelaher, M. (2017). Canadian school-based HPV vaccine programs and policy considerations. *Vaccine*, *35*(42), 5700–5707. https://doi.org/10.1016/j.vaccine.2017.07.079

76. Chang M. H. (2014). Prevention of hepatitis B virus infection and liver cancer. *Recent results in cancer research. Fortschritte der Krebsforschung. Progrès dans les recherches sur le cancer*, *193*, 75–95. https://doi.org/10.1007/978-3-642-38965-8_5

77. Hepatitis B. Immunize BC. (2022, May 27). Retrieved July 27, 2022, from https://immunizebc.ca/hepatitis-b

78. Kids Boost Immunity. (n.d.). Retrieved July 27, 2022, from https://kidsboostimmunity.com/

Appendix A

Table A1

Estimated percentages of incident cancer cases in Canadians (2015) attributable to established risk factors

Exposure	Cancer Site(s)	Incident Cases
Tobacco Smoking	Lung, Colorectal, Liver, Oral Cavity & Pharynx, Larynx, Esophagus, Stomach, Pancreas, Kidney, Bladder, Breast, Cervix	17.5%
Physical Inactivity	Lung, Colorectal, Esophagus, Stomach, Kidney, Bladder, Breast	4.9%
Sedentary Behaviour	Colorectal, Breast	1.7%
Excess Weight	Colorectal, Liver, Esophagus, Stomach, Pancreas, Kidney, Breast, Prostate	3.1%
Alcohol Consumption	Colorectal, Liver, Oral Cavity & Pharynx, Larynx, Esophagus, Stomach, Pancreas, Breast	1.8%
Low Vegetable Consumption, Red meat Consumption	Colorectal	0.3%, 0.6%
Processed meat consumption	Colorectal, Stomach	0.6%
Ultraviolet Radiation	Melanoma	2.3%
Outdoor Air Pollution ($PM_{2.5}$), Residential Radon	Lung	0.9%, 0.9%
Hepatitis B virus	Liver	0.1%
Human papillomavirus	Oral Cavity & Pharynx, Larynx, Cervix	2.0%
All exposures (including those not shown)	**Lung, Colorectal, Liver, Oral cavity & Pharynx, Larynx, Esophagus, Melanoma, Stomach, Non-Hodgkin's lymphoma, Pancreas, Kidney, Bladder, Breast, Cervix, Prostate**	**33.3%**

Source: Poirier et al., 2019

Appendix B

Table B1

ECLIPSE search strategy question prompts and answers

Expectation	What is being identified, improved and/or evaluated?	The information found will inform public health programs and policies aimed at reducing cancer risk factors and exposures in B.C.
Client Group	Who is the service being provided for?	The service is for individuals living without cancer in B.C.
Location	Where is the service run from?	The service will be provided by local health authorities and public health organizations in B.C.
Impact	What is the desired change to service, if any? How is success measured?	The desired change in service is that public health professionals will be able to link cancer risk indicators to public health goals for cancer prevention, in order to better inform current and future programs and policies. Success of existing programs is measured by effectiveness in reducing modifiable risk based on evidence and data through evaluation.
Professionals	Who is involved in providing the service?	Professionals providing the service include public health professionals, health care providers, policy makers and workplace managers
SErvice	What is the type of service?	The service that is being researched is primary prevention of cancer for risk factor X

Table B2

ECLIPSE search terms used for environmental scan and literature searches on Google, Google Scholar and PubMed

Concept 1: **E**xpectation		Concept 2: **C**lient Group / Concept 3: **L**ocation		Concept 4: **P**rofessionals		Concept 5: **SE**rvice		Cancer Risk Factor	Keywords
policy	AND	"British Columbia"	AND	"public health"	AND	cancer	AND	Smoking	smoking OR smoke* OR "Smoking cessation" OR Tobacco*
policies		BC				prevention		Alcohol	alcohol*
program*		B.C.				"cancer prevention"		Physical Activity	"physical activity" OR exercise
		Canada				"primary prevention"		Diet	diet* OR "healthy eating" OR nutrition
						risk		UV Radiation	UV OR "UV radiation" OR "ultraviolet radiation" OR sun OR "sun exposure"
						"cancer risk"		Environmental	environment* OR radon OR "air pollution" OR "air quality" OR "particulate matter"
								Occupational	occupation* OR worker*
								HPV	HPV OR human papillomavirus OR "cervical cancer"
								Hepatitis B	"hepatitis B" OR "hep B"

Appendix C

Table C1
Screening of resources from environmental scan and literature review

Risk Factor	Environmental Scan	Literature Review	
	Grey Literature (Internal, External), n =	Title/ Abstract Screen, n =	Full-Text Screen, n =
Smoking	19	23	11
Alcohol	11	16	7
Physical Activity	19	19	7
Diet	18	12	6
UV Radiation	9	13	8
Environmental	23	14	9
Occupational	15	18	10
Infections	12	16	9

Appendix D

Table D1

Attributable cancer cases due to diet in Canada for all ages in 2015

	Low Fruit Consumption	Low Vegetable Consumption	Processed Meat Consumption	Red Meat Consumption	Excess Weight (BMI)
Canada	7.3%	4.7%	3.8%	5.5%	5.6%
B.C.	7.5%	3.9%	3.7%	5.6%	4.6%

Source: ComPARe, 2015

Appendix E

Figure E1

Prevalence of Occupational Exposure Estimates in British Columbia

Carcinogen	Low Exposure (# of individuals)	Moderate Exposure (# of individuals)	High Exposure (# of individuals)	Total Exposed (# of individuals)
Solar Radiation	47,802	64,054	126,043	237,899
Shift Work	NA	NA	NA	220,913
Gasoline Engine Exhaust	140,091	45,383	12,577	198,051
Diesel Engine Exhaust	113,040	14,977	2,060	130,077
Polycyclic Aromatic Hydrocarbons	NA	NA	NA	67,925
Silica, Crystalline	17,152	3,286	36,595	57,033
Wood Dust	14,871	28,872	12,678	56,421
Second Hand Smoke	NA	NA	NA	56,045
Benzene	39,071	2,420	81	41,572
Welding Fumes	5,652	12,465	19,829	37,946
Asbestos	16,988	16,123	1,663	34,774
Lead and Lead Compounds, Inorganic	19,565	8,355	5,329	33,249

Note. Estimates for shift work are for 2011

Source: CAREX Canada, 2016